

Higher Professional Training Course in Cybersecurity

2024/2025

English Applied to Computer Science(18117)

1st year | 1st semester

Lesson 3 and 4 (25th September 2025)

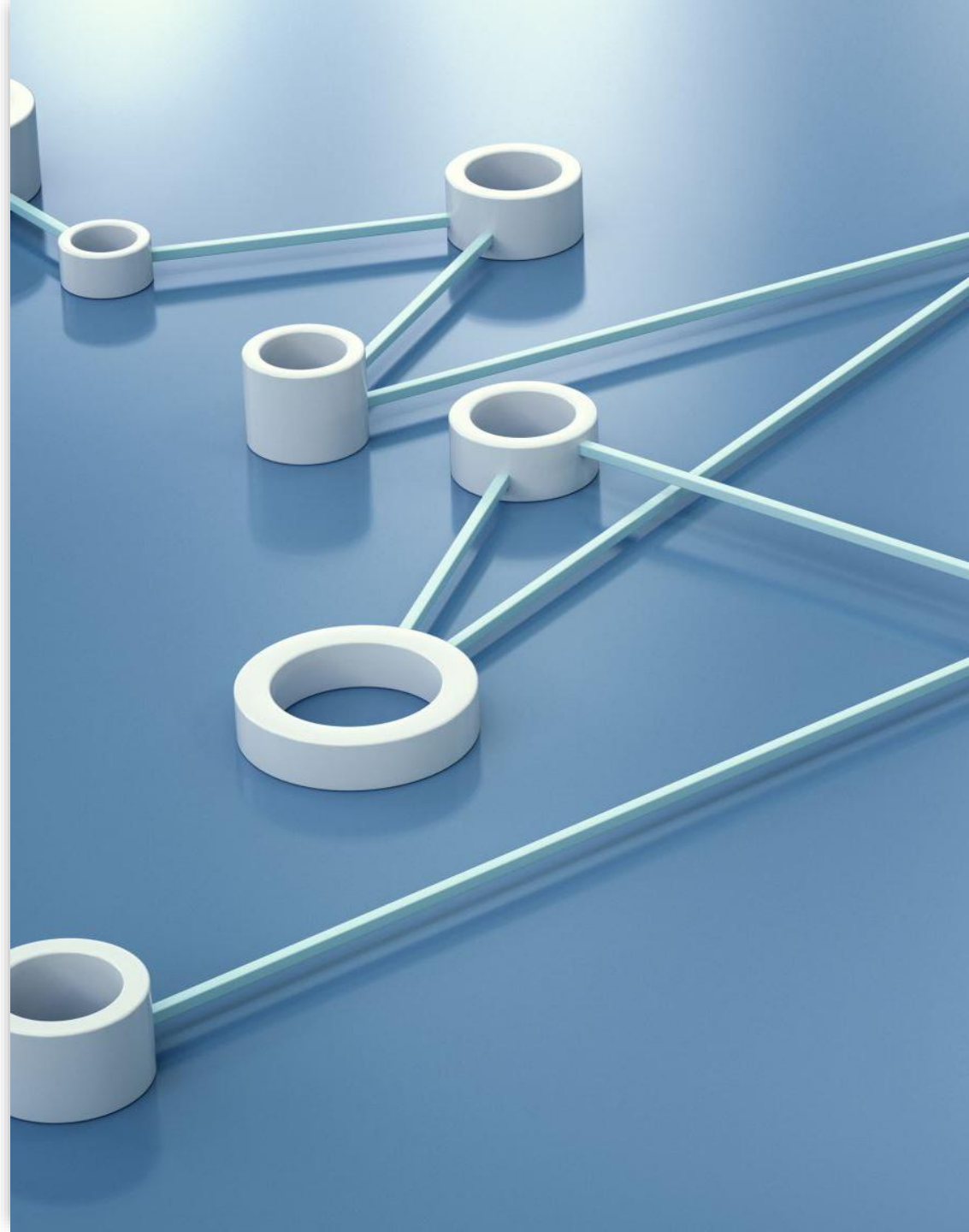


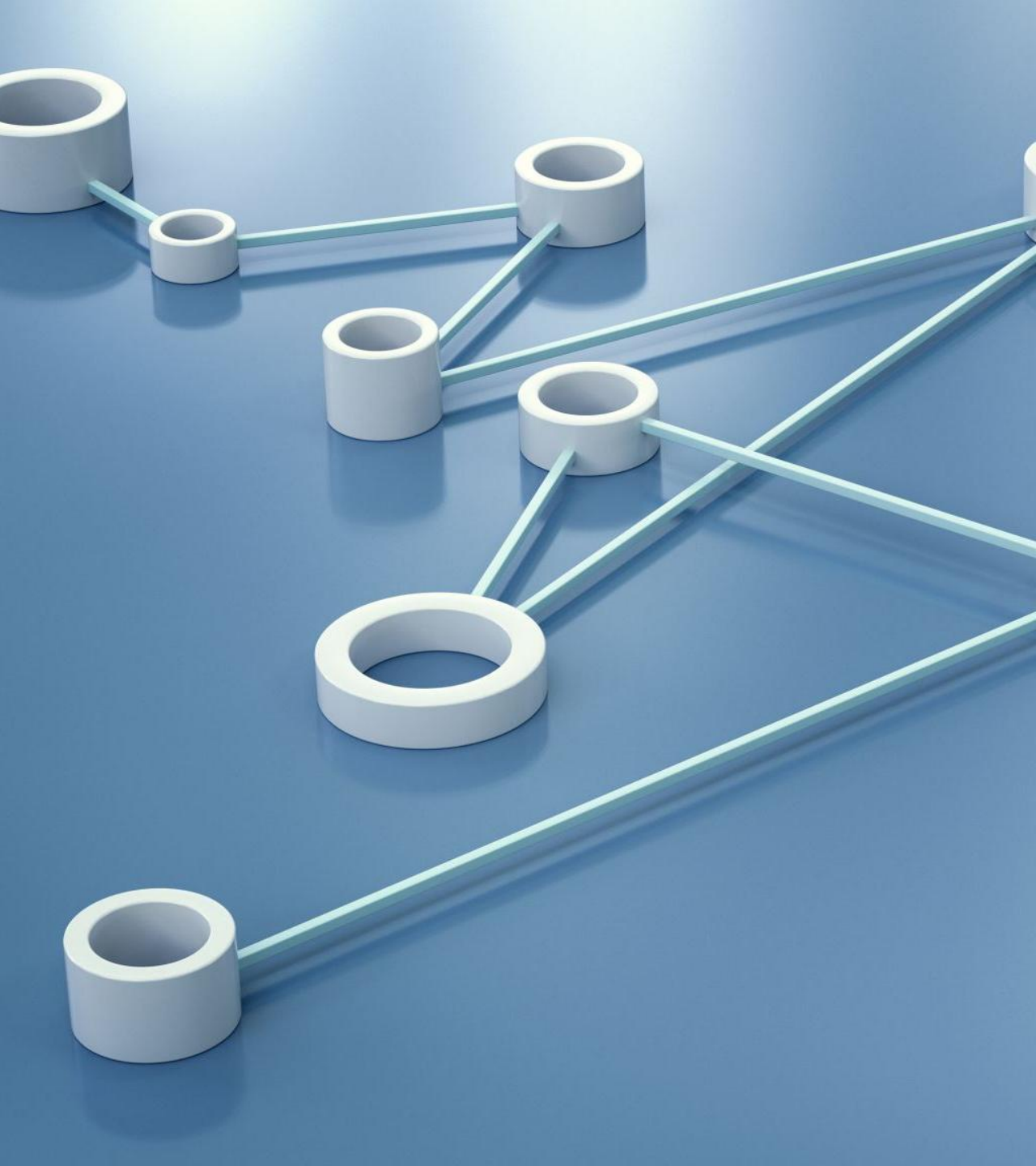


User Interfaces

What are user interfaces (UI)?

The user interface (UI) is the point of human-computer interaction and communication in a device. This can include display screens, keyboards, a mouse and the appearance of a desktop. It is also the way through which a user interacts with an application or a website.



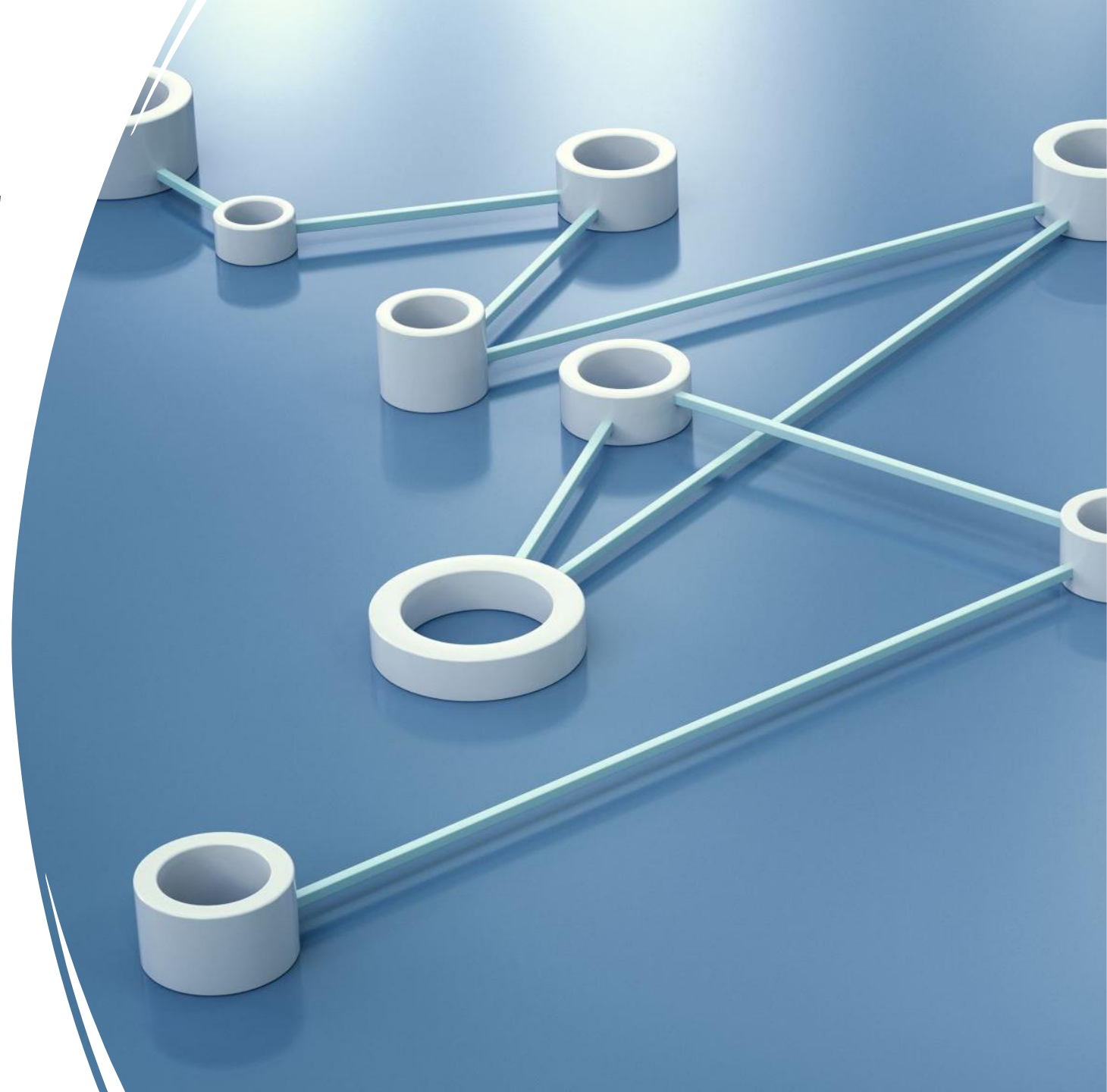


Types of user interfaces (UI)?

- graphical user interface (GUI)
- command line interface (CLI)
- menu-driven user interface
- touch user interface
- voice user interface (VUI)
- form-based user interface
- natural language user interface

Examples of user interfaces (UI)?

- computer mouse
- remote control
- virtual reality
- ATMs
- speedometer
- the old iPod click wheel



Start

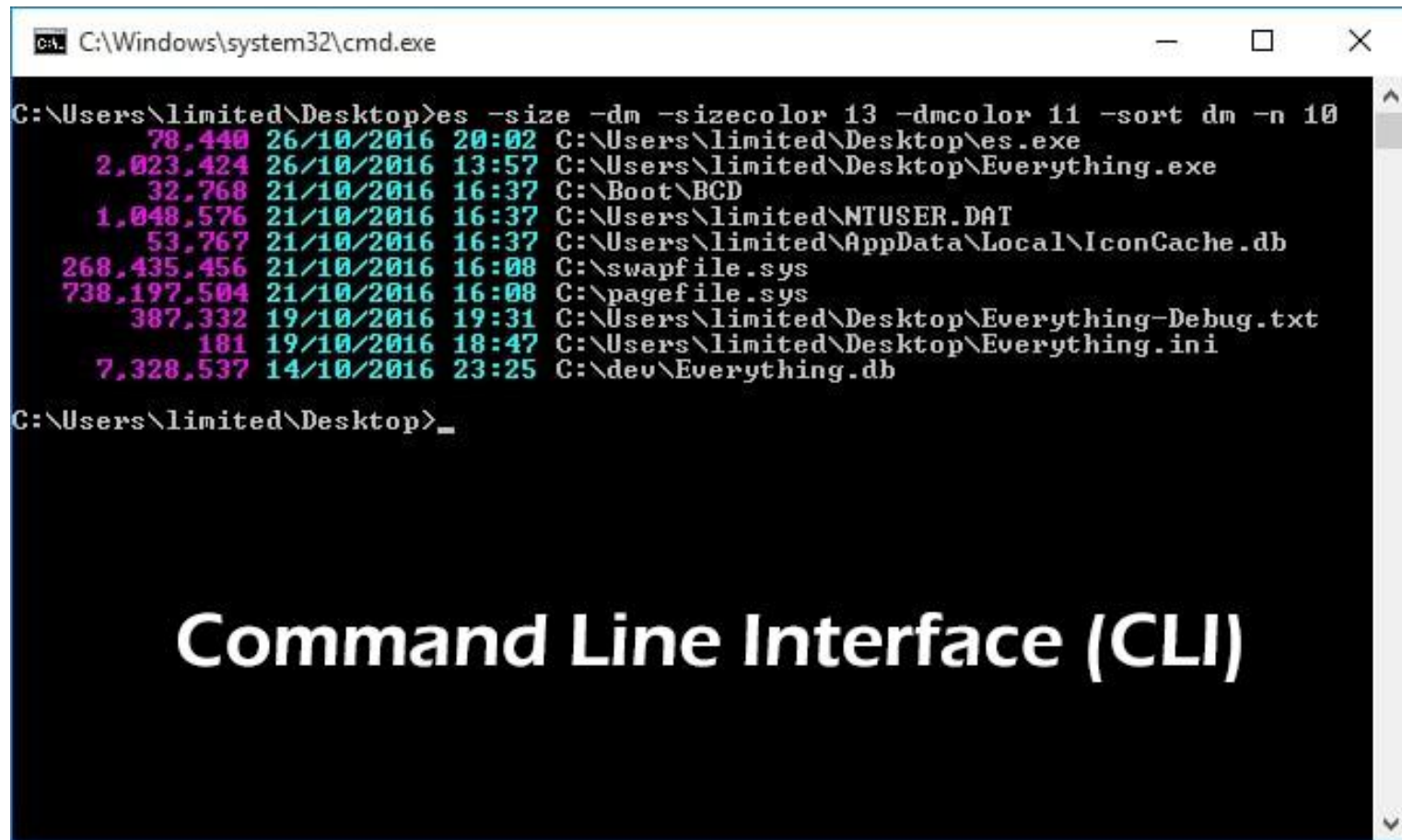
User 



Graphical User Interface (GUI /gu:i:/)

Elements of a GUI include such things as windows, pull-down menus, buttons, scroll bars and icons. With the increasing use of multimedia as part of the GUI, sound, voice, motion video and virtual reality are increasingly becoming the GUI for many applications.





```
C:\Windows\system32\cmd.exe

C:\Users\limited\Desktop>es -size -dm -sizecolor 13 -dmcolor 11 -sort dm -n 10
 78,440 26/10/2016 20:02 C:\Users\limited\Desktop\es.exe
2,023,424 26/10/2016 13:57 C:\Users\limited\Desktop\Everything.exe
 32,768 21/10/2016 16:37 C:\Boot\BCD
1,048,576 21/10/2016 16:37 C:\Users\limited\NTUSER.DAT
 53,767 21/10/2016 16:37 C:\Users\limited\AppData\Local\IconCache.db
268,435,456 21/10/2016 16:08 C:\swapfile.sys
738,197,504 21/10/2016 16:08 C:\pagefile.sys
387,332 19/10/2016 19:31 C:\Users\limited\Desktop\Everything-Debug.txt
 181 19/10/2016 18:47 C:\Users\limited\Desktop\Everything.ini
7,328,537 14/10/2016 23:25 C:\dev\Everything.db

C:\Users\limited\Desktop>_
```

Command Line Interface (CLI)

A **text-based** user interface used to **view and manage computer files**. Command line interfaces are also called command-line user interfaces, console user interfaces and character user interfaces.

Advantages of GUI

- ❑ Easy to use due to the visual nature of the interface.
- ❑ The user has lots of visual clues as to what things do.
- ❑ Less chance of user input errors.
- ❑ Quicker to input commands.

Disadvantages of GUI

- ❑ To display all of the nice graphics required by a GUI takes a lot of computing power so quite a powerful computer is needed.
- ❑ User restricted to pre-determined options.

Advantages of CLI

- ❑ The User has direct communication with computer.
- ❑ Has a wider range of commands to use.

Disadvantages of CLI

- ❑ User must remember complex commands.
- ❑ Lots of typing needed for quite simple tasks.

— Booklet page 11 | ex. 5.1

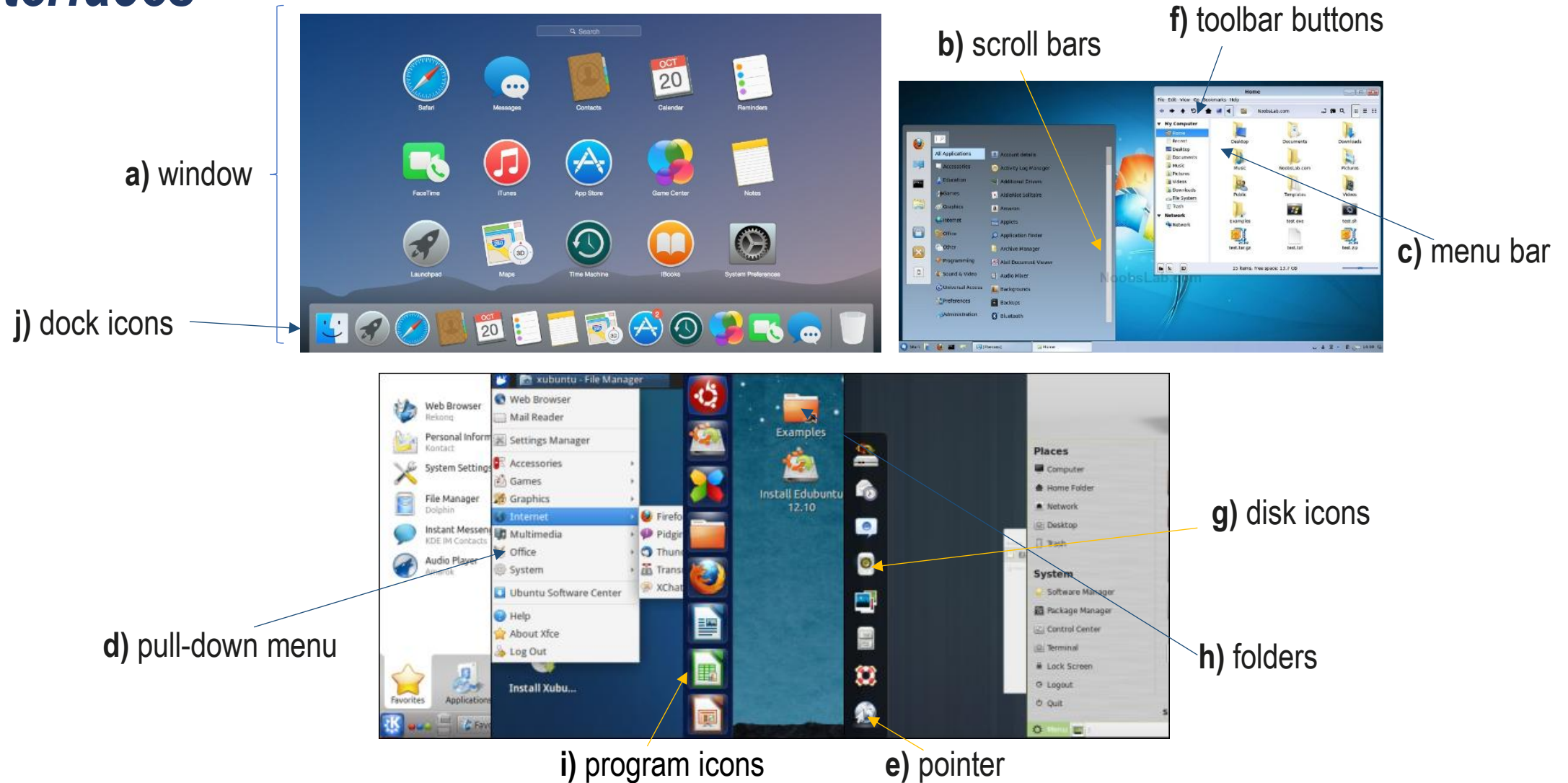
5.1. Match the interface elements with the following definitions.

- | | |
|--------------------|--|
| a) window | <u> f </u> a menu that the user 'pulls down' from a name in the menu bar at the top of the screen by selecting the name with the mouse |
| b) dock | <u> e </u> graphic images (or intuitive symbols) used to represent an object or task |
| c) toolbar buttons | <u> d </u> containers for documents and applications |
| d) folders | <u> c </u> found at the top of a window, they take you to the Home folder and others. |
| e) icons | <u> b </u> set of icons at the bottom of the screen that give you instant access to the things you use most |
| f) pull-down menu | <u> g </u> an arrow, controlled by the mouse, that allows you to move around the screen |
| g) the pointer | <u> a </u> a viewing area less than or equal to the screen size. By using different windows, you can work on several documents or applications simultaneously. |



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ex. 5.2

User Interfaces



User Interfaces

WIMP

displays
self-explanatory
documents
programs
arrow
options
drop-down/ pull down
desktop
user-friendly

A GUI makes use of a **WIMP** environment: **Windows**, **Icons**, **Menus** and **Pointer**.

This type of interface is _____.

System functions are accessed by selecting _____ **icons** (pictures representing _____ or _____) and items from menus (_____).

The **pointer** is the _____ controlled by the mouse, which lets you choose _____ from menus.

The background screen that _____ icons, representing programs, files and folders (directories) is called the _____.

User Interfaces

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The background screen that displays icons, representing programs, files and folders (directories) is called the desktop.

User Interfaces

GUI operations

-
- | | |
|----------------------------------|---------------------------------------|
| 1. Double click on the title bar | a) to open a new menu. |
| 2. Click on a menu | b) and the window fills the screen. |
| 3. Right-click on an icon | c) if you want to move the window. |
| 4. Slide the scrollbar down | d) to hide the window. |
| 5. Click the 'Minimise' button | e) to scroll the window down. |
| 6. Drag de title bar | f) to open it. |
| 7. Select the icon | g) and its background changes colour. |

6. Read the Starting Mailbag Deluxe manual and answer the questions below.

Starting Mailbag Deluxe

Follow these instructions to start Mailbag Deluxe after installation. This process applies to all standard GUIs.

- a. Find the shortcut to Mailbag Deluxe on your desktop. The icon is a folder with a star in the corner.
- b. Right-click the icon to show the dropdown menu.
- c. Select the option that says "Open Mailbag".
- d. The setup guide will open. Click "Next".
- e. Type your name and email address. Make sure the cursor appears in the correct box.
- f. Click "Finish Setup". This step may take a few moments.
- g. Click "Run Mailbag". You are ready to use Mailbag Deluxe!

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6.1. Mark the following statements as True (T) or False (F).

- | | |
|---|--------------|
| a) The instructions help users install a program. | <u> T </u> |
| b) "Open Mailbag" is an option from the dropdown menu. | <u> T </u> |
| c) The cursor takes a few moments to appear in the correct box. | <u> F </u> |

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6.2. Choose which word or phrase best fits each blank.

icon | user | organise | dropdown menu

- a) A(n) user operates a computer.
- b) A(n) icon can represent a file, program, or folder.
- c) A(n) Dropdown menu provides a list of options when clicked on by a user.
- d) organise a desktop by arranging the icons on it.

6.3. Identify the sentences that use the words correctly.

- a) The user clicked on the cursor to start the web browser. ✗ icon
- b) Some people display pictures on their desktops. ✓
- c) Clicking on a menu option will often run a program. ✓
- d) Use the icon to scroll down to the bottom of the page. ✗ bar

7. Read the manual and answer the questions below.

GUIs

The term *user interface* refers to the standard procedures the user follows to interact with a particular computer. A few years ago, the way in which users had access to a computer system was quite complex. They had to memorize and type a lot of *commands* just to see the content of a disk, to copy files or to respond to a single prompt. In fact, only experts used computers, so there was no need for a user-friendly interface. Now, however, computers are used by all kinds of people and as a result there is a growing emphasis on the user interface.

A good user interface is important because when you buy a program you want to use it easily. Moreover, a graphical user interface saves a lot of time: you don't need to memorize commands in order to execute an application; you only have to point and click so that its content appears on the screen.

Macintosh computers — with a user interface based on graphics and intuitive tools — were designed with a single clear aim: to facilitate interaction with the computer. This interface is called **WIMP: Window, Icon, Menu (or Mouse) and Pointer** and software products for the Macintosh have been designed to take full advantage of its features using this interface. In addition, the ROM chips of a Macintosh contain libraries that provide program developers with routines for generating windows, dialogue boxes, icons and pop-up menus. This ensures the creation of applications with a high level of consistency.

Today, the most innovative GUIs are the Macintosh, Microsoft Windows and IBM OS/2 Warp. These three platforms include similar features: a desktop with icons, windows and folders, a printer selector, a file finder, a control panel and various desk accessories. Double-clicking a folder opens a window which contains programs, documents or further nested folders. At any time within a folder, you can launch the desired program or document by double-clicking the icon, or you can drag it to another location.

The three platforms differ in other areas such as device installation, network connectivity or compatibility with application programs.

These interfaces have been so successful because they are extremely easy to use. It is well known that computers running under an attractive interface stimulate users to be more creative and produce high quality results, which has a major impact on the general public.

- a) What is the contribution of Macintosh computers to the development of graphic environments?
- b) What does the acronym 'WIMP' stand for?
- c) What computing environments based on graphics are mentioned in the text?
- d) How do *you* run a program *on* a computer with a graphical interface?

a) The creation of the WIMP interface; ROM (read-only memory) chips that contain libraries that provide routines (= consistency).

b) WIMP = Window, Icon, Menu/Mouse, Pointer (user-friendly).

c) Macintosh, Microsoft Windows and IBM OS/2 Warp.

d) You can launch the desired program by double-clicking.



Grammar review

Verbs + object + infinitive or to-infinitive

Allow, enable and permit are used with this structure:

Verb + object + to-infinitive

Let is used with this structure:

Verb + object + infinitive

Help can be used with either structure.

Study the examples

A GUI **lets you point** to icons and click a mouse button to execute a task.

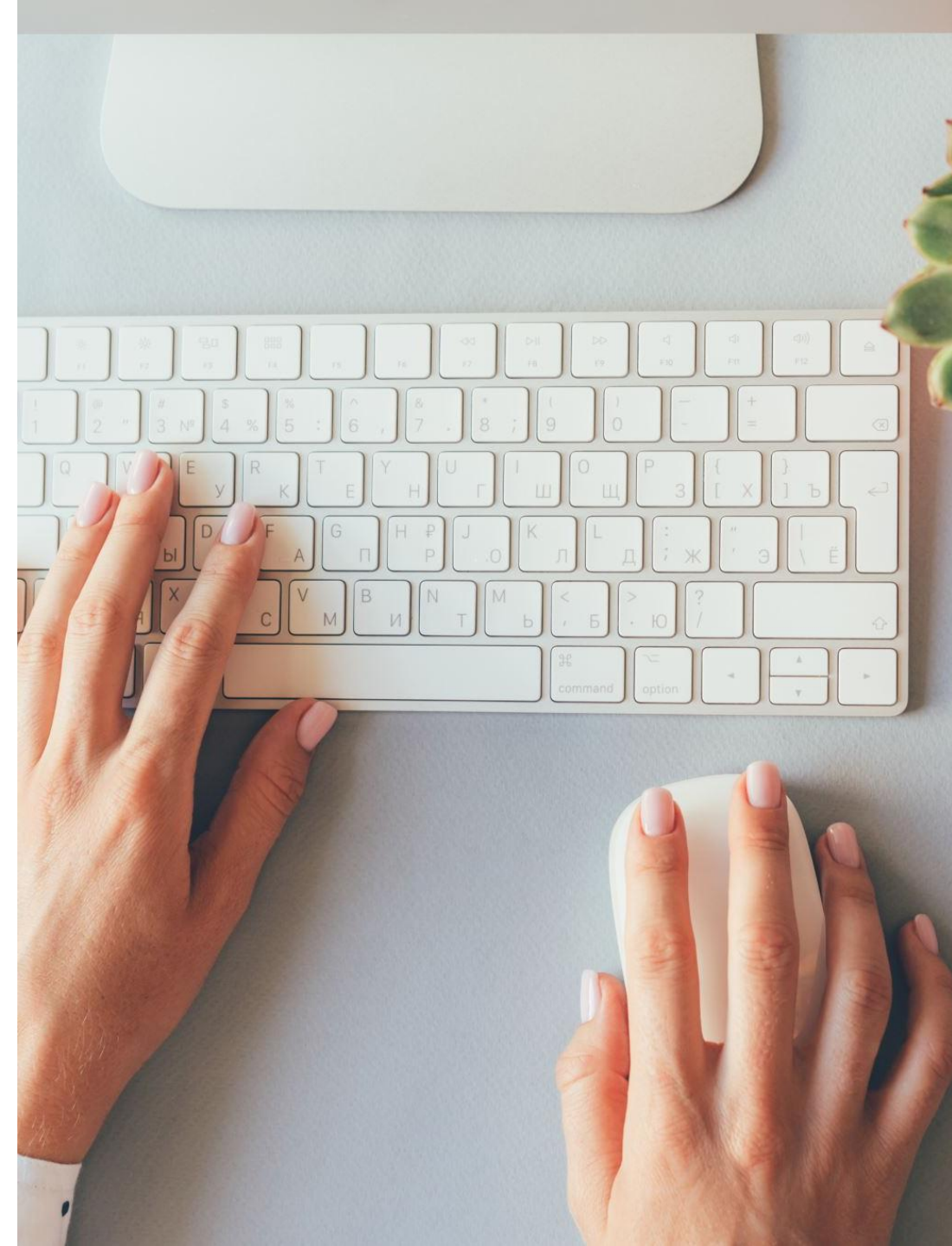
A GUI **allows you to** use a computer without knowing any operating system commands.

The X Window System **enables Unix-based computers to** have a graphical look and feel.

Voice recognition software **helps disabled users (to)** access computers.

8) Fill in the gaps with the correct form of the verb.

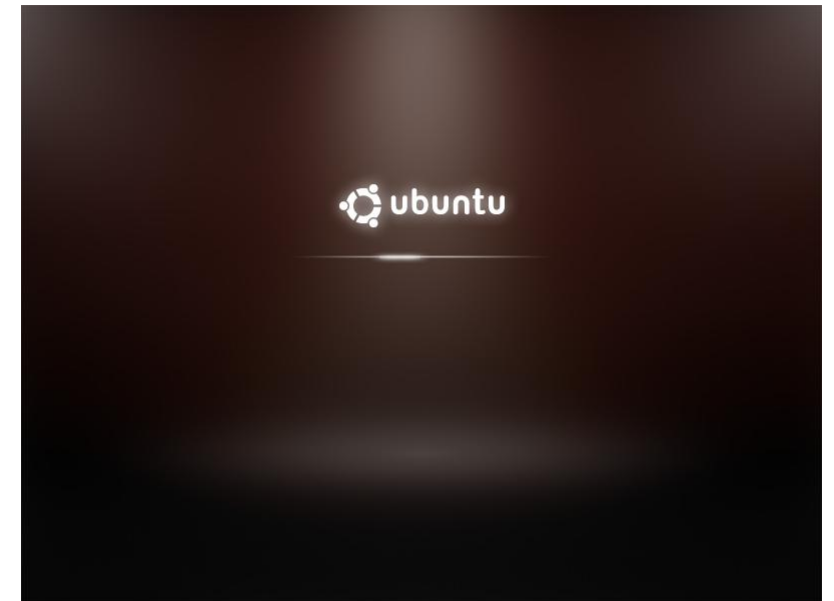
- a) Adding more memory lets your computer
— — (work) faster.
- b) The MouseKeys feature enables you
(use) the numeric keypad to move the mouse pointer.
- c) ALT + TAB allows you (switch) between
programs.
- d) ALT + PRINT SCREEN lets you (copy) an
image of an active window to the Clipboard.
- e) The StickyKeys feature helps disabled people
 (operate) two keys simultaneously.



Mobile UI Design (Booklet, page 14, ex. 9)



Splash screen



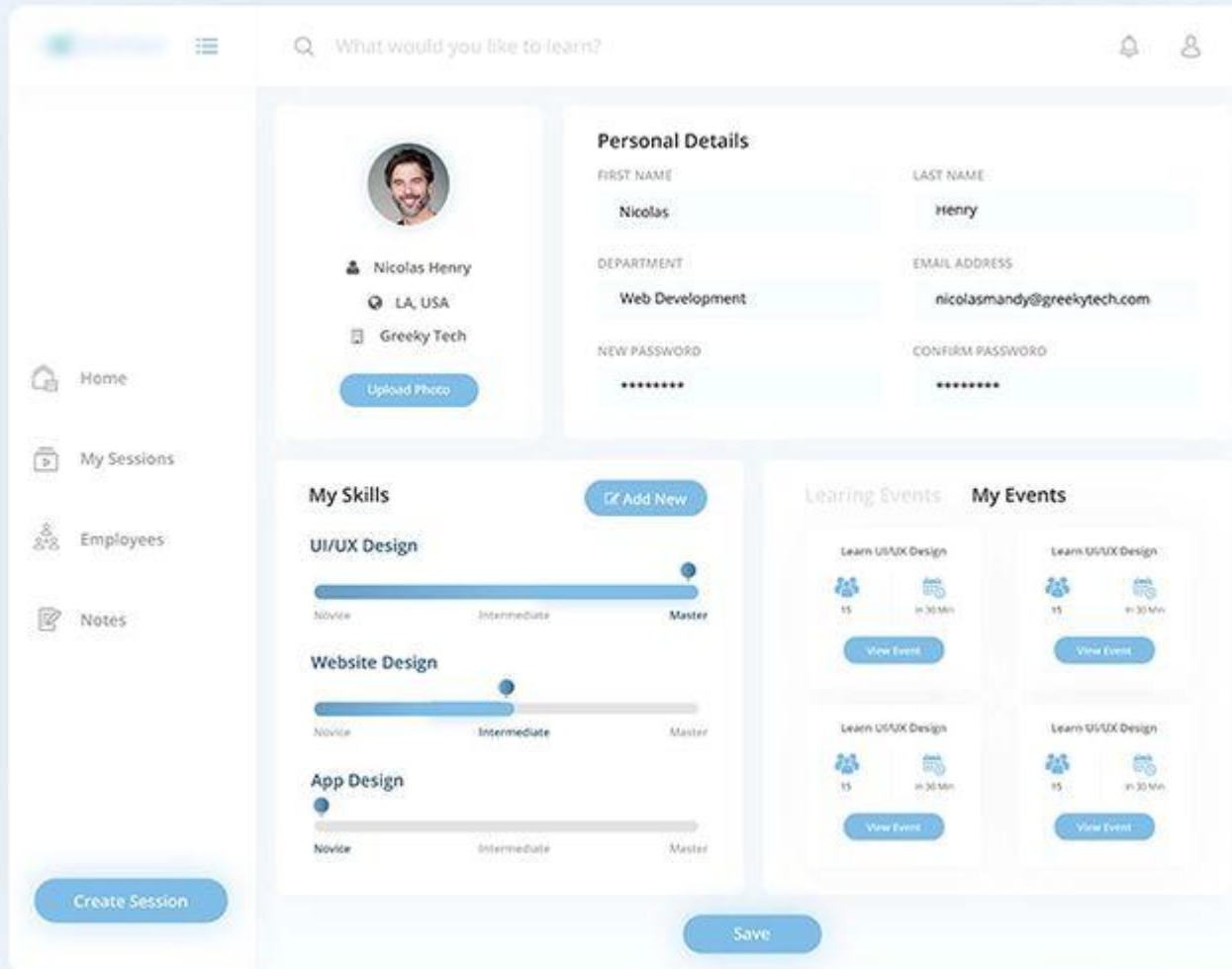
Mobile UI Design



Home screen



Mobile UI Design



A mobile application profile screen design. The interface includes a top navigation bar with a search icon and the text "What would you like to learn?", a bell icon, and a user icon. A left sidebar contains navigation links: Home, My Sessions, Employees, and Notes. The main content area is divided into several sections: a user profile card for "Nicolas Henry" with location "LA, USA" and company "Greeky Tech", an "Upload Photo" button, and a "Personal Details" form with fields for First Name, Last Name, Department, Email Address, New Password, and Confirm Password. Below this is a "My Skills" section with progress bars for "UI/UX Design", "Website Design", and "App Design", each with "Novice", "Intermediate", and "Master" levels. To the right is a "Learning Events" section with a "My Events" tab and four event cards, each titled "Learn UI/UX Design" with a "15" icon, a "30 Min" duration, and a "View Event" button. At the bottom left is a "Create Session" button, and at the bottom center is a "Save" button.

What would you like to learn?

Home
My Sessions
Employees
Notes

Create Session

Save

Personal Details

FIRST NAME: Nicolas
LAST NAME: Henry
DEPARTMENT: Web Development
EMAIL ADDRESS: nicolasmandy@greekytech.com
NEW PASSWORD: *****
CONFIRM PASSWORD: *****

My Skills [Add New](#)

UI/UX Design
Novice Intermediate Master

Website Design
Novice Intermediate Master

App Design
Novice Intermediate Master

Learning Events **My Events**

Learn UI/UX Design
15 30 Min
[View Event](#)

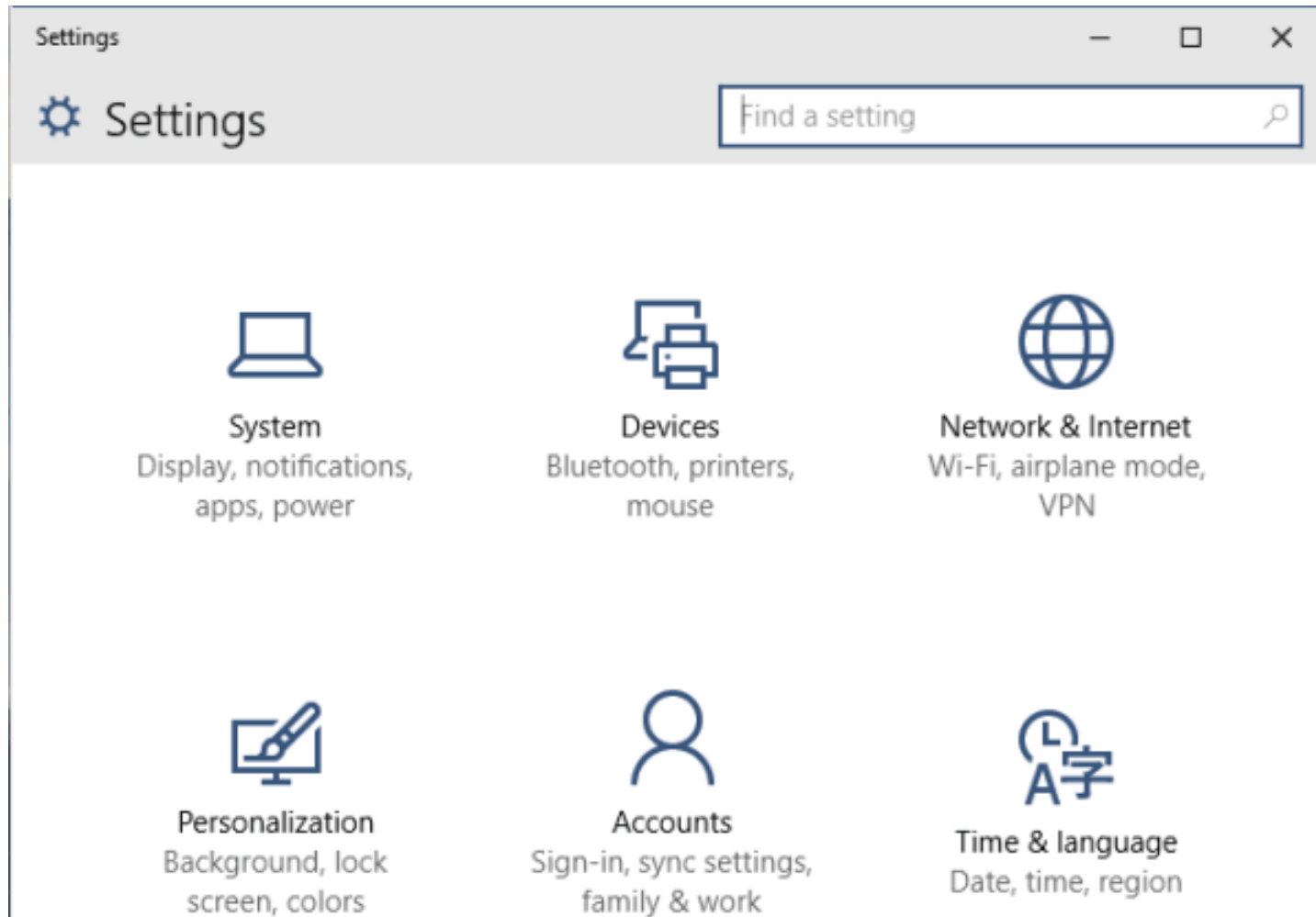
Learn UI/UX Design
15 30 Min
[View Event](#)

Learn UI/UX Design
15 30 Min
[View Event](#)

Learn UI/UX Design
15 30 Min
[View Event](#)

Profile screen

Mobile UI Design



Settings screen

Mobile UI Design

Onboarding Screen

- 5 Screen
- Easy Editable
- Fully Vector



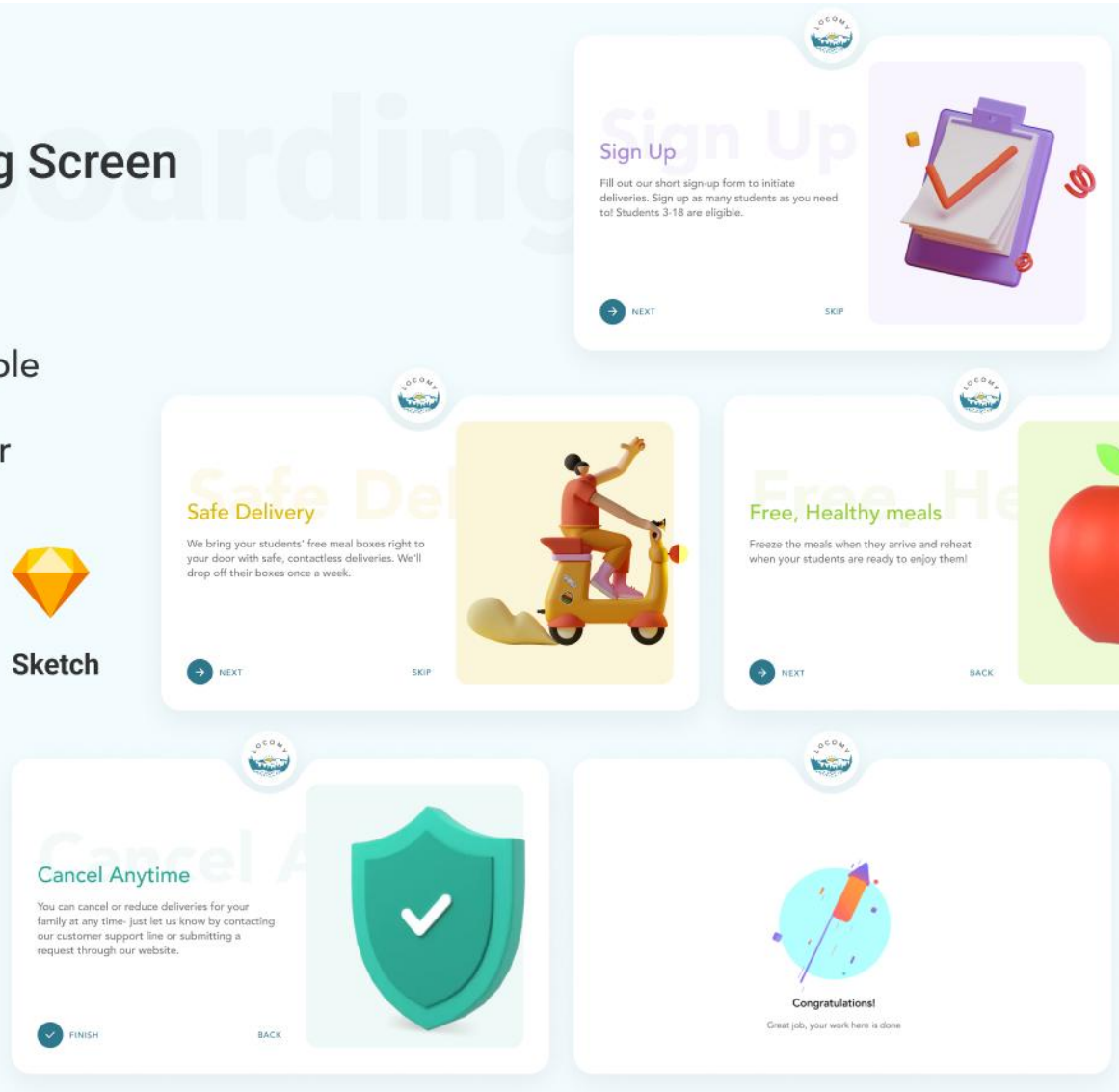
Figma



XD

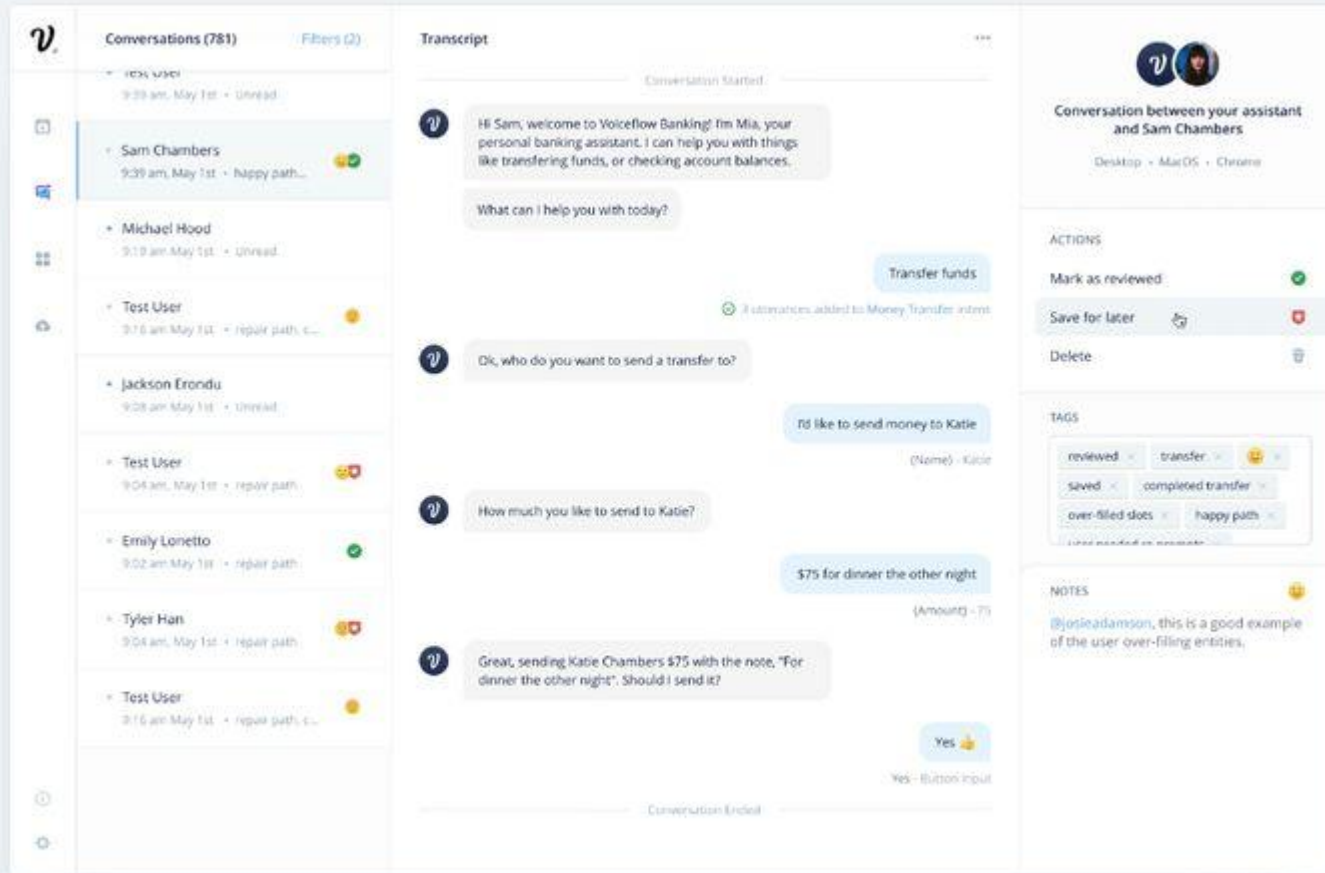


Sketch



Onboarding screen

Mobile UI Design



Conversational screen